

7. (a) A pendulum oscillating with simple harmonic motion consists of a small 0.05 kg mass oscillating at the end of a string of length 4.0 m. The displacement x in metres at time t can be written as:

$$x = 0.25 \cos \omega t$$

- (i) Show that the angular velocity, ω , is approximately 1.57 rad s^{-1} . [2]

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- (ii) Calculate the maximum speed of the mass. [2]

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- (iii) Show that the kinetic energy (E_k) of the mass, in joules, may be written as: [2]

$$E_k = 3.8 \times 10^{-3} \sin^2 1.57t$$

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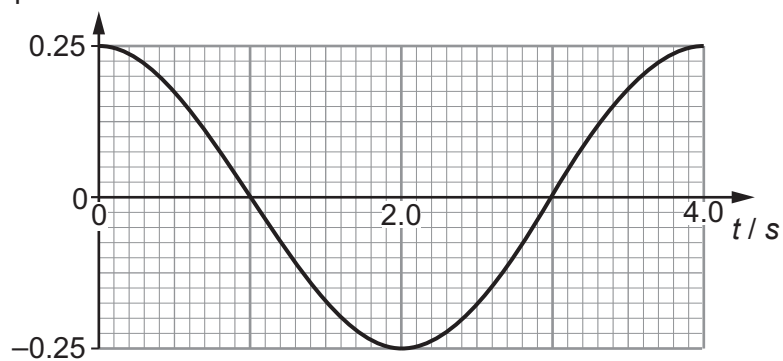
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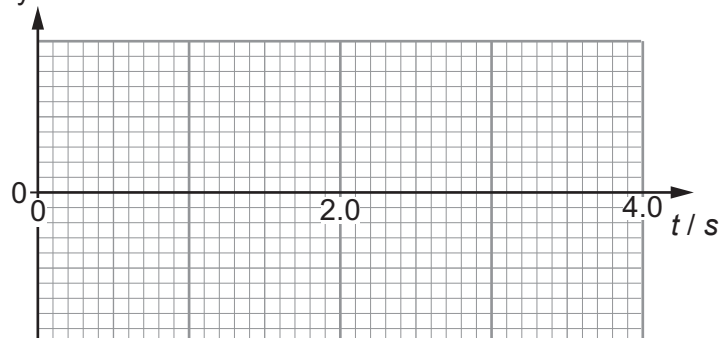
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- (iv) The displacement-time curve for the pendulum is shown for a time interval of one period i.e. 4.0 s. Draw curves for velocity, v , and E_k on the axes below for the same interval of time. Indicate values on your axes for v and E_k . [4]

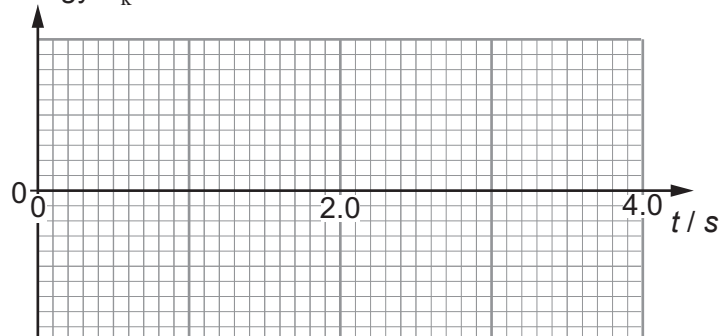
Displacement x / m



Velocity v / ms^{-1}



Kinetic energy E_k / J



- (v) Determine the maximum height reached by the pendulum mass above the level of its lowest point. [2]

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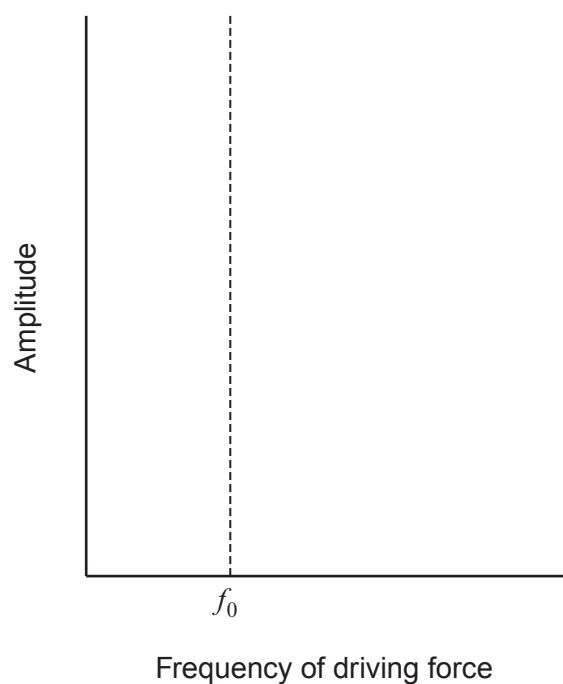
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(b) A system that can oscillate may be driven by an external sinusoidal force.

- (i) Sketch the amplitude of a lightly damped system as the frequency of the driver is increased. The natural frequency f_0 of the system is indicated. Label the curve **X**.

[1]



- (ii) Name the effect seen near the frequency, f_0 .

[1]

- (iii) Sketch a second curve on the **axes above** for an identical oscillating system that is damped to a greater extent. Label the curve as **Y**.

[2]